PART A — QUESTIONS

AF17-001
Question:
Solve: $19x - 13 = 101$
A) 5
B) 6
C) 7
D) 8

AF17-002
Question:
If $17n + 5 = 124$, what is n?
A) 5
B) 6
C) 7
D) 8

AF17-003
Question:
Simplify: $4(5x - 6) + 7x$
A) $20x - 24$
B) $27x - 24$
C) $27x - 6$
D) $13x - 24$

AF17-004
Question:
A company charges $30 per delivery plus a $22 scheduling fee. Which expression represents the total cost for d deliveries?
A) $30d + 22$
B) $22d + 30$
C) $52d$
D) $30 - 22d$

AF17-005
Question:
If $f(x) = 14x - 31$, what is $f(8)$?
A) 71
B) 81
C) 112
D) 143

AF17-006
Question:
Which ordered pair satisfies the equation $y = 12x - 5$?
A) (1, 8)
B) (2, 20)
C) (3, 31)
D) (4, 42)

AF17-007
Question:
Solve: $18(x - 4) = 126$
A) 9
B) 10
C) 11
D) 12

AF17-008
Question:
A pattern begins: 13, 29, 45, 61, ... What is the next number?
A) 75
B) 76
C) 77
D) 78

AF17-009
Question:
Solve: $x/28 = 4$
A) 32
B) 84
C) 112
D) 124

AF17-010
Question:
A line has a slope of -14 and crosses the y-axis at 11. Which equation represents the line?
A) $y = 11x - 14$
B) $y = -14x + 11$
C) $y = 14x + 11$
D) $y = -11x + 14$

AF17-011
Question:
If 8 boxes contain 232 parts, how many parts are in 6 boxes at the same rate?
A) 156
B) 168
C) 174
D) 184

AF17-012
Question:
Solve: $180 - 22x = 92$
A) 3
B) 4
C) 5
D) 6

AF17-013
Question:
Which expression is equivalent to $70p - 31p + 8$?

A) 39p + 8
B) 101p + 8
C) 39p - 8
D) 70p - 23

AF17-014
Question:
A storage area starts with 960 units. Workers add 45 units each day. Which equation gives the total number of units U after d days?
A) U = 960d + 45
B) U = 45d + 960
C) U = 1,005d
D) U = 960 - 45d

AF17-015
Question:
If y = -16x + 200, what is y when x = 11?
A) 24
B) 34
C) 176
D) 376

AF17-016
Question:
Solve: 28x - 35 = 18x + 65
A) 8
B) 9
C) 10
D) 11

AF17-017
Question:
Which value of x makes 15x - 9 greater than 126?
A) 8
B) 9
C) 10
D) 7

AF17-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: -8, 9, 26, 43

What is the rule?
A) y = 17x - 8
B) y = 8x + 17
C) y = -8x + 17
D) y = 17x + 8

AF17-019
Question:
Solve: 14(x + 5) = 11x + 103
A) 9
B) 10
C) 11
D) 12

AF17-020
Question:
If a = -15 and b = 8, what is 2a + 4b?
A) -62
B) -2
C) 2
D) 62

AF17-021
Question:
Which point is located in Quadrant IV?
A) (13, -7)
B) (-13, 7)
C) (-13, -7)
D) (13, 7)

AF17-022
Question:
A worker counts 180 items in 15 minutes. At the same rate, how many items can the worker count in 8 minutes?
A) 84
B) 90
C) 96
D) 108

AF17-023
Question:
Solve for z: A = 24z - 48
A) z = A + 48
B) z = (A + 48)/24
C) z = 24(A + 48)
D) z = A/24 - 48

AF17-024
Question:
What is the slope of a line that goes through (5, 9) and (11, 69)?
A) 8
B) 9
C) 10
D) 60

AF17-025
Question:

Simplify: $52x - 23 - 41x + 31$
A) $11x + 8$
B) $93x + 8$
C) $11x - 54$
D) $93x - 54$

AF17-026
Question:
If $g(x) = x^2 - 10x$, what is $g(14)$?
A) 42
B) 56
C) 140
D) 196

AF17-027
Question:
A sequence follows the rule “add 12, then multiply by 3.” If the first term is 6, what is the second term?
A) 18
B) 36
C) 54
D) 66

AF17-028
Question:
Solve: $18(x - 5) + 14 = 140$
A) 10
B) 11
C) 12
D) 13

AF17-029
Question:
A line crosses the y-axis at 18 and falls 9 units for every 1 unit to the right. Which equation represents the line?
A) $y = 9x + 18$
B) $y = -9x + 18$
C) $y = 18x - 9$
D) $y = -18x + 9$

AF17-030
Question:
If $22/33 = x/99$, what is x?
A) 44
B) 55
C) 66
D) 77

AF17-031
Question:
Solve: $-24x + 72 = -168$

A) -10
B) -6
C) 6
D) 10

AF17-032
Question:
A stockroom starts with 3,000 connectors. Each machine uses 150 connectors. Which expression gives the number of connectors left after m machines?
A) 3,000 + 150m
B) 3,000 - 150m
C) 150m - 3,000
D) 3,000 ÷ 150m

AF17-033
Question:
Which equation has x = 22 as its solution?
A) 2x + 13 = 56
B) 5x - 24 = 86
C) x/11 + 9 = 12
D) 3x - 12 = 52

PART B — ANSWER KEY + EXPLANATIONS

AF17-001 — Correct: B — Explanation:
Solve 19x - 13 = 101. Add 13 to both sides: 19x = 114. Divide by 19: x = 6. The most common mistake is subtracting 13 instead of adding 13.

AF17-002 — Correct: C — Explanation:
Start with 17n + 5 = 124. Subtract 5 from both sides: 17n = 119. Divide by 17: n = 7. A common mistake is stopping at 119 and forgetting to divide by 17.

AF17-003 — Correct: B — Explanation:
Distribute first: 4(5x - 6) = 20x - 24. Then add 7x: 20x + 7x - 24 = 27x - 24. A common mistake is forgetting to multiply -6 by 4.

AF17-004 — Correct: A — Explanation:
The company charges $30 for each delivery, so that part is 30d. The $22 scheduling fee is added one time. The total cost is 30d + 22. A common mistake is treating the scheduling fee as the per-delivery charge.

AF17-005 — Correct: B — Explanation:
Substitute 8 for x: f(8) = 14(8) - 31 = 112 - 31 = 81. A common mistake is adding 31 instead of subtracting it.

AF17-006 — Correct: C — Explanation:
Test the choices in y = 12x - 5. For (3, 31), 12(3) - 5 = 36 - 5 = 31, which matches the y-value. A common mistake is choosing a point that is close but does not make the equation true.

AF17-007 — Correct: C — Explanation:
Solve $18(x - 4) = 126$. Divide both sides by 18: $x - 4 = 7$. Add 4: $x = 11$. A common mistake is adding 4 before undoing the multiplication by 18.

AF17-008 — Correct: C — Explanation:
The pattern increases by 16 each time: 13, 29, 45, 61. Add 16 to 61: 77. A common mistake is assuming the pattern increases by 15 because the numbers are close to multiples of 15.

AF17-009 — Correct: C — Explanation:
$x/28 = 4$ means x divided by 28 equals 4. Multiply both sides by 28: $x = 112$. A common mistake is adding 28 and 4 instead of multiplying.

AF17-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -14 and the y-intercept is 11, so the equation is $y = -14x + 11$. A common mistake is switching the slope and y-intercept.

AF17-011 — Correct: C — Explanation:
Find the unit rate: 232 parts ÷ 8 boxes = 29 parts per box. For 6 boxes: $29 \times 6 = 174$. A common mistake is multiplying 232 by 6 without finding the rate first.

AF17-012 — Correct: B — Explanation:
Solve $180 - 22x = 92$. Subtract 180 from both sides: $-22x = -88$. Divide by -22: $x = 4$. A common mistake is losing the negative sign after subtracting 180.

AF17-013 — Correct: A — Explanation:
Combine like terms: $70p - 31p = 39p$. The constant 8 stays the same. The expression becomes $39p + 8$. A common mistake is combining the constant with the variable terms.

AF17-014 — Correct: B — Explanation:
The storage area starts with 960 units. Workers add 45 units each day. After d days, the total is $U = 45d + 960$. A common mistake is making 960 the daily rate.

AF17-015 — Correct: A — Explanation:
Substitute $x = 11$ into $y = -16x + 200$: $y = -16(11) + 200 = -176 + 200 = 24$. A common mistake is treating $-16(11)$ as positive 176.

AF17-016 — Correct: C — Explanation:
Solve $28x - 35 = 18x + 65$. Subtract 18x: $10x - 35 = 65$. Add 35: $10x = 100$. Divide by 10: $x = 10$. A common mistake is adding 35 to only one side.

AF17-017 — Correct: C — Explanation:
Solve $15x - 9 > 126$. Add 9: $15x > 135$. Divide by 15: $x > 9$. The only listed value greater than 9 is 10. A common mistake is choosing 9, but $15(9) - 9 = 126$, not greater than 126.

AF17-018 — Correct: A — Explanation:
The y-values increase by 17 each time x increases by 1, so the slope is 17. When $x = 0$, $y = -8$, so the y-intercept is -8. The rule is $y = 17x - 8$. A common mistake is using +8 instead of -8.

AF17-019 — Correct: C — Explanation:
Distribute: $14x + 70 = 11x + 103$. Subtract 11x: $3x + 70 = 103$. Subtract 70: $3x = 33$. Divide by 3: $x = 11$. A common mistake is forgetting to multiply 5 by 14.

AF17-020 — Correct: C — Explanation:
Substitute a = -15 and b = 8: 2a + 4b = 2(-15) + 4(8) = -30 + 32 = 2. A common mistake is treating 2(-15) as positive 30.

AF17-021 — Correct: A — Explanation:
Quadrant IV has a positive x-coordinate and a negative y-coordinate. The point (13, -7) fits this description. A common mistake is choosing (-13, 7), which is Quadrant II.

AF17-022 — Correct: C — Explanation:
180 items in 15 minutes means 180 ÷ 15 = 12 items per minute. In 8 minutes: 12 × 8 = 96. A common mistake is multiplying 180 by 8 without finding the rate first.

AF17-023 — Correct: B — Explanation:
Start with A = 24z - 48. Add 48 to both sides: A + 48 = 24z. Divide by 24: z = (A + 48)/24. A common mistake is dividing A by 24 first and then adding 48.

AF17-024 — Correct: C — Explanation:
Slope equals change in y divided by change in x. From (5, 9) to (11, 69), y increases by 60 and x increases by 6. The slope is 60 ÷ 6 = 10. A common mistake is using 60 as the slope without dividing by the change in x.

AF17-025 — Correct: A — Explanation:
Combine like terms: 52x - 41x = 11x, and -23 + 31 = 8. The simplified expression is 11x + 8. A common mistake is writing 11x - 8 because the positive 31 is missed.

AF17-026 — Correct: B — Explanation:
Substitute 14 for x: g(14) = 14^2 - 10(14) = 196 - 140 = 56. A common mistake is calculating 14^2 as 28 instead of 196.

AF17-027 — Correct: C — Explanation:
Start with 6. Add 12: 18. Then multiply by 3: 54. A common mistake is multiplying first and then adding 12.

AF17-028 — Correct: C — Explanation:
Solve 18(x - 5) + 14 = 140. Distribute: 18x - 90 + 14 = 140. Combine: 18x - 76 = 140. Add 76: 18x = 216. Divide by 18: x = 12. A common mistake is forgetting to combine -90 and +14.

AF17-029 — Correct: B — Explanation:
“Falls 9 units for every 1 unit to the right” means the slope is -9. The line crosses the y-axis at 18, so the equation is y = -9x + 18. A common mistake is using a positive slope because the number 9 is given.

AF17-030 — Correct: C — Explanation:
Use the proportion 22/33 = x/99. Since 33 × 3 = 99, multiply 22 by 3: x = 66. A common mistake is adding 66 to the numerator because 33 becomes 99.

AF17-031 — Correct: D — Explanation:
Solve -24x + 72 = -168. Subtract 72 from both sides: -24x = -240. Divide by -24: x = 10. A common mistake is giving -10 because both sides contain negative numbers.

AF17-032 — Correct: B — Explanation:
The stockroom starts with 3,000 connectors. Each machine uses 150 connectors, so m machines use 150m connectors. The number left is 3,000 - 150m. A common mistake is adding 150m even though connectors are being used.

AF17-033 — Correct: B — Explanation:
Test x = 22. In choice B, 5(22) - 24 = 110 - 24 = 86, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.